The Poisson Distribution

ANSWER KEY

Section 1: The Poisson Probability Mass Function

1 If $X \sim \text{Po}(\lambda)$, what is the probability mass function $P(X = k)$ for $k = 0, 1, 2, \ldots$?

a. $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$ ✓

b. $P(X = k) = \frac{\lambda^k e^{\lambda}}{k!}$

c. $P(X = k) = \binom{n}{k}\lambda^k(1-\lambda)^{n-k}$

d. $P(X = k) = \lambda e^{-\lambda k}$

The Poisson probability mass function is given by $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$ for $k \in \{0, 1, 2, \ldots\}$.

2 A call center receives an average of 3 calls per minute ($X \sim \text{Po}(3)$). What is the probability of receiving exactly 2 calls in a given minute?

a. $\frac{9}{2}e^{-3} \approx 0.224$ ✓

b. $3e^{-3}$

c. $\frac{27}{6}e^{-3}$

d. e^{-3}

$P(X = 2) = \frac{e^{-3}3^2}{2!} = \frac{9}{2}e^{-3} = 4.5e^{-3} \approx 0.2240.$

3 For $X \sim \text{Po}(\lambda)$, what is $P(X = 0)$?

a. $e^{-\lambda}$ ✓

b. 0

c. 1

d. $\lambda e^{-\lambda}$

Substituting $k = 0$: $P(X = 0) = \frac{e^{-\lambda}\lambda^0}{0!} = \frac{e^{-\lambda} \cdot 1}{1} = e^{-\lambda}$.

Section 2: Mean, Variance, and Poisson Characteristics

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4 What is a distinctive mathematical property of any Poisson random variable $X \sim \text{Po}(\lambda)$?

a. **The mean and variance are strictly equal:** $E(X) = \text{Var}(X) = \lambda$ ✓

b. The mean is always λ^2

c. The standard deviation equals λ

d. The median always equals $\lambda + 1$

A defining feature of the Poisson distribution is that its expected value and variance are both equal to the rate parameter λ.

5 If a Poisson random variable has variance $\text{Var}(X) = 16$, what is its standard deviation?

a. 4 ✓

b. 16

c. 256

d. $\sqrt{4} = 2$

Standard deviation is $\sigma = \sqrt{\text{Var}(X)} = \sqrt{16} = 4$.

6 Under what practical conditions does a Poisson process serve as a valid model?

a. **Events occur independently, singly, and at a constant average rate across continuous time or space** ✓

b. There are exactly two trials with equal probability

c. Events must occur in clusters at predetermined times

d. The probability of an event increases as time elapses

A Poisson process assumes events happen independently at a constant average rate, without simultaneous multiple arrivals.

Section 3: Changing Time Intervals & Sum of Independent Poisson RVs

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7 Flaws on a wire occur at an average rate of 1.5 per meter. What is the distribution of flaws Y in a 4-meter segment of wire?

a. $Y \sim \text{Po}(6)$ ✓

b. $Y \sim \text{Po}(1.5)$

c. $Y \sim \text{Po}(4)$

d. $Y \sim \text{Po}(9)$

Poisson rates scale linearly with interval size: $\lambda_{\text{new}} = 1.5 \times 4 = 6$. Thus $Y \sim \text{Po}(6)$.

8 If $X \sim \text{Po}(4)$ and $Y \sim \text{Po}(7)$ are independent random variables, what is the distribution of $W = X + Y$?

a. $W \sim \text{Po}(11)$ ✓

b. $W \sim \text{Po}(28)$

c. $W \sim \text{Po}(3)$

d. $W \sim N(11, 28)$

The sum of independent Poisson random variables is itself a Poisson random variable whose parameter is the sum of the individual rates: $\lambda_W = 4 + 7 = 11$.

9 If $W \sim \text{Po}(11)$, what is the variance of W?

a. 11 ✓

b. 121

c. $\sqrt{11}$

d. 22

For any Poisson random variable $W \sim \text{Po}(\lambda)$, $\text{Var}(W) = \lambda = 11$.

Section 4: Poisson Approximation to the Binomial Distribution

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10 When can a Binomial distribution $B(n, p)$ be well approximated by a Poisson distribution $\text{Po}(\lambda)$?

a. **When n is very large ($n > 50$) and p is very small ($p < 0.1$), with $\lambda = np$** ✓

b. When n is small and $p = 0.5$

c. When $np \geq 10$ and $n(1-p) \geq 10$

d. For any binomial distribution where $n > 30$

The Poisson distribution is the limiting form of the Binomial distribution as $n \to \infty$ and $p \to 0$ such that $np = \lambda$ remains moderate.

11 A factory produces microchips with defect probability $p = 0.002$. In a batch of $n = 1000$ chips, approximate the probability of finding zero defects.

a. $e^{-2} \approx 0.135$ ✓

b. $e^{-0.002}$

c. $1 - e^{-2}$

d. 0.002

$\lambda = np = 1000(0.002) = 2$. $P(X = 0) \approx e^{-\lambda} = e^{-2} \approx 0.1353$.

12 If $X \sim \text{Po}(\lambda)$ has $P(X = 1) = P(X = 2)$, find the value of λ.

a. $\lambda = 2$ ✓

b. $\lambda = 1$

c. $\lambda = 0.5$

d. $\lambda = 4$

$\frac{e^{-\lambda}\lambda^1}{1!} = \frac{e^{-\lambda}\lambda^2}{2!} \Rightarrow \lambda = \frac{\lambda^2}{2} \Rightarrow 1 = \frac{\lambda}{2} \Rightarrow \lambda = 2$.

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